A certify-or-reject layer for GENCO under N-1 contingencies

First results on GENCO's published Texas 2000-bus N-1 predictions.

Data: huggingface.co/gridfm/genco-pf-contingency (paper checkpoint)

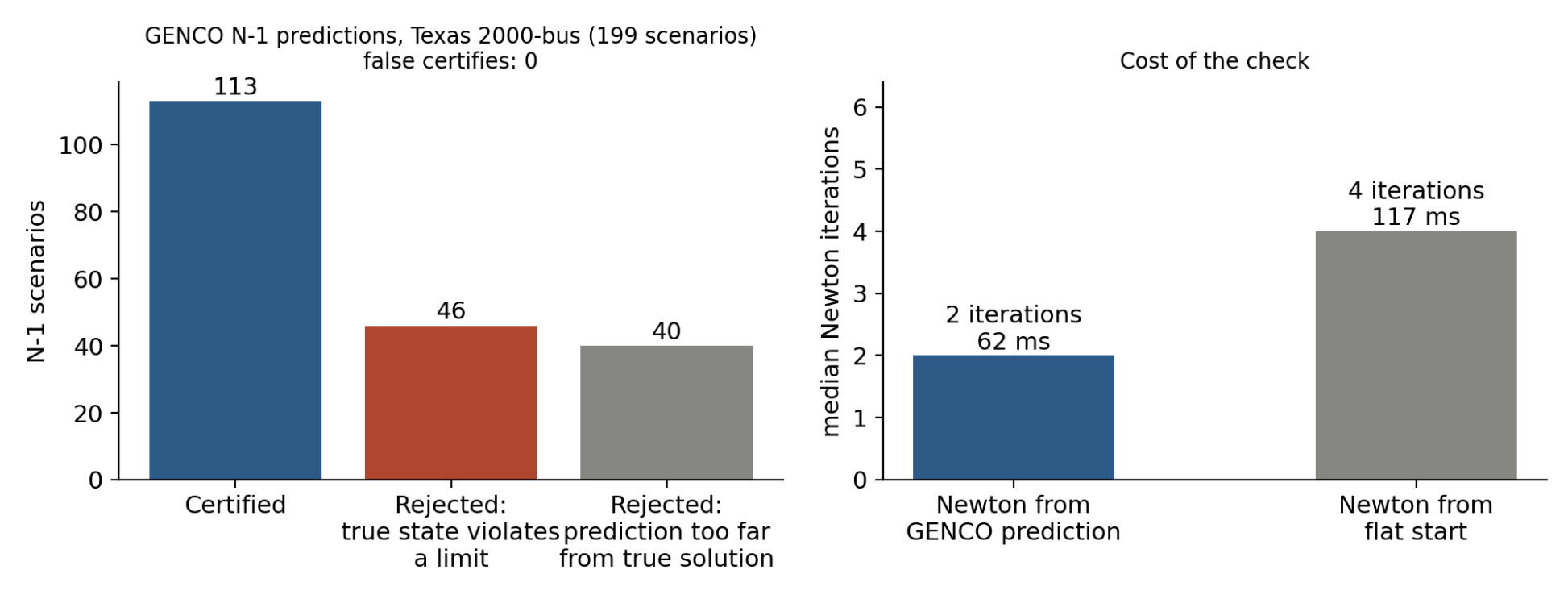
GENCO drives power-balance residuals down, but nothing guarantees limit satisfaction.

- For PF there is no box projection, so voltage, thermal and generator reactive limits are not enforced by the model.
- Raw predictions also can't be limit-checked directly. Evaluated as-is, 189 of 199 N-1 predictions fail a limit check, mostly generator reactive limits (52 of 60 on a breakdown vs 16 of 60 for the true states), because reactive power recovered from an approximate state is sensitive to small voltage errors.
- A downstream user needs one bit per scenario they can trust: is this an AC-feasible, limit-satisfying operating point?

Solve from the prediction, then check.

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1. Newton-Raphson warm-started from GENCO's (Vm, θ); PV/ref set points fixed; converge to 1e-8 pu
2. Distance: converged point within 0.01 pu and 1° of the prediction at every bus
3. Limits at the converged point: bus voltage bounds, branch MVA ratings (both ends),
   bus-aggregated generator Q limits
4. CERTIFIED only if 1-3 pass; otherwise REJECTED with the reason, and the case goes to a full solve
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- Model-agnostic, plain numpy/scipy. Every certified verdict audited against the published ground truth.
- Sanity check first: the published ground truth satisfies the rebuilt power-flow equations to about 1e-9 pu on every scenario.



- **113 certified, 0 false certifies.** All 59 scenarios whose true N-1 state violates a limit were rejected.
- 27 feasible scenarios were rejected only because the prediction was more than 0.01 pu / 1° away. That share depends on the tolerance, which is a design choice.

What this does not show yet, and how it could plug in.

- **Coverage:** 199 of 8,823 N-1 scenarios. The other partitions and the N-2 to N-20 splits use the same pipeline.
- **Checks not yet included:** angle-difference limits and per-generator Q limits (they're aggregated per bus now).
- **Integration:** a thin adapter so `gridfm_graphkit predict` outputs pass straight into `certify()`, reporting a verdict, a reason and a margin per scenario.
- **Open question for the team:** which tolerance and which limits would be most useful for your users, and is the Texas contingency set the right first benchmark?